F18CD Mark Scheme - December 2024

1. (a) Let A be a non-empty set of real numbers that is bounded below. Then the infimum of A is the greatest lower bound of A. ✓✓✓

(b)

Proof. By the Archimedean property ✓, there exists an $n \in \mathbb{N}$ such that

$$\frac{1}{n} < y - x.✓$$

By definition of $\mathbb{N} \subset \mathbb{R}$, there exists a smallest element $m \in S$ with

$$S = \{k \in \mathbb{N} : k > nx\}.✓✓$$

Hence, $nx < m$. Since $m - 1 \notin S$, ✓this means that $m - 1 \leq nx \Rightarrow m \leq nx + 1 < ny$ ✓. Putting all this together gives

$$x < \frac{m}{n} < y.✓$$

□

(c)

Proof. Since $1 \in \mathbb{Q}$ then $1 \in A$ and hence $A \neq \emptyset$ ✓. By the definition of A, every element $a \in A$ is less than 3 and greater than or equal to 1. Thus 1 is a lower bound for A and 3 is an upper bound ✓. By part (b), for every $\varepsilon > 0$, there exists a $a, b \in A$ such that

$$a < 1 + \varepsilon✓$$

and

$$3 - \varepsilon < b.✓$$

Thus $\inf(A) = 1$ and $\sup(A) = 3$. Since $1 \in A$ it follows that $\min(A) = 1$ ✓. However, as $3 \notin A$ it follows that the maximum of A does not exist ✓. □

(d) The minimum does always exist.

Proof. Since $B \subset \mathbb{R}$ is bounded below, $\inf(B)$ exists ✓. Let $\inf(B) = \beta$. Suppose towards a contradiction that $\beta \notin B$. Then there exists a $b \in B$ such that

$$\beta < b < \beta + 1.✓$$

Again, since $\beta \notin B$, there exists a $c \in B$ such that

$$c < b.✓$$

Now, it cannot be that

$$b - 1 < c < b$$

and hence $c \leq b - 1 < \beta$. Contradiction. ✓ □

2.(a) A sequence (a_n) converges to $a \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon$$

for every $n > N$. ✓✓✓

(b)

Proof. For every $\varepsilon > 0$ there exists an $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon ✓$$

for every $n > N_1$. Similarly, there exists an $N_2 \in \mathbb{N}$ such that

$$|c_n - a| < \varepsilon ✓$$

for every $n > N_2$. Define $N = \max(N_1, N_2, m)$ ✓. Then it follows that for every $n > N$ ✓, we have

$$\begin{aligned} a - \varepsilon < a_n \leq b_n \leq c_n < a + \varepsilon ✓ \\ \Rightarrow |b_n - a| < \varepsilon. ✓ \end{aligned}$$

Hence $\lim(b_n) = a$. □

(c)

Proof. As $-1 \leq \cos(n^3) \leq 1$ for all $n \in \mathbb{N}$ ✓it follows that

$$-\frac{1}{n^5} \leq x_n \leq \frac{1}{n^5} ✓$$

for all $n \in \mathbb{N}$. It follows by the algebraic limit theorem that $\lim(-1/n^5) = \lim(1/n^5) = 0$ and hence by part (b) it follows that $\lim(x_n) = 0$ ✓. □

Proof. Since $n^n > 0$ and $n! > 0$ for all $n \in \mathbb{N}$ it follows that $y_n \geq 0$ for all $n \in \mathbb{N}$ ✓. Now,

$$\begin{aligned} \frac{n!}{n^n} &= \frac{n}{n}\frac{n-1}{n}\frac{n-2}{n}\cdots\frac{2}{n}\frac{1}{n} \\ &= 1\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\frac{2}{n}\frac{1}{n} ✓ \end{aligned}$$

Now $0 < 1 - k/n < 1$ for each $k \in \mathbb{N}$ with $0 < k < n - 1$ and all $n \in \mathbb{N}$ ✓. Hence,

$$\frac{n!}{n^n} \leq \frac{1}{n}$$

for all $n \in \mathbb{N}$. That is,

$$0 \leq y_n \leq \frac{1}{n} ✓$$

for all $n \in \mathbb{N}$. Since $\lim(1/n) = 0$ it follows by part (b) that $\lim(y_n) = 0$. ✓ □

(d) No, a counterexample is $a_n = 0$, $b_n = n$ and $c_n = n^2$ ✓✓. Then $a_n \leq b_n \leq c_n$ for all $n \in \mathbb{N}$ but $\lim(b_n)$ does not exist. ✓

3.(a) A series $\sum_{k=1}^{\infty} a_k$ is absolutely convergent if $\sum_{k=1}^{\infty} |a_k|$ is convergent. ✓✓✓

(b)

Proof. The difference of the sequence of partial sums with $n > m$ is given by

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n.$$ ✓

($\Rightarrow$) If (s_n) is convergent, it follows that it is Cauchy and hence for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \varepsilon$$

for every $n > m > N$. ✓✓

($\Leftarrow$) If for every $\varepsilon > 0$ there exists a $N \in \mathbb{N}$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for every $n > m > N$ it follows that (s_n) is Cauchy ✓✓. By the Cauchy criterion for sequences, it follows that (s_n) is convergent. ✓

□

(c)

Proof. Let

$$a_k = \frac{3k^2 + \cos^3(k)}{k^7}$$

with $k \in \mathbb{N}$. Then since $\cos^3(k) \leq 1$ for all $k \in \mathbb{N}$ ✓it follows that

$$\begin{aligned} |a_k| &\leq \frac{3}{k^5} + \left|\frac{\cos^3(k)}{k^7}\right| \\ &\leq \frac{3}{k^5} + \frac{1}{k^7}. \end{aligned}$$ ✓

By the P-series test, the absolute convergence test and algebraic limit theorem of series it follows that

$$\sum_{k=1}^{\infty} \frac{3}{k^5} + \frac{1}{k^7}$$

is convergent ✓✓. Hence, by the comparison test it follows that

$$\sum_{k=1}^{\infty} \frac{3k^2 + \cos^3(k)}{k^7}$$

is convergent. ✓ □

(d)

Proof. Between $0 < x < \pi/2$ the function $\sin(x)$ is strictly increasing ✓. Since $\dfrac{1}{k} < \pi/2$ for all $k \in \mathbb{N}$ it follows that

$$\sin\left(\frac{1}{k+1}\right) < \sin\left(\frac{1}{k}\right).$$

That is, the sequence (a_k) defined by $a_k = \sin(1/k)$ is monotone decreasing ✓. Since $\sin(x)$ is continuous everywhere, it follows that

$$\begin{aligned}\lim_{k\to\infty}\left(\sin\left(\frac{1}{k}\right)\right) &= \sin\left(\lim_{k\to\infty}\left(\frac{1}{k}\right)\right)\\ &= \sin(0)\\ &= 0.\checkmark\end{aligned}$$

Hence by the alternating series test it follows that

$$\sum_{k=1}^{\infty}(-1)^k \sin\left(\frac{1}{k}\right)$$

is convergent. ✓ □

It doesn't follow that the series

$$\sum_{k=1}^{\infty}(-1)^k \cos\left(\frac{1}{k}\right)$$

is convergent. This series is divergent. It follows since $\cos(x)$ is continuous that

$$\begin{aligned}\lim_{k\to\infty}\left(\cos\left(\frac{1}{k}\right)\right) &= \cos\left(\lim_{k\to\infty}\left(\frac{1}{k}\right)\right)\\ &= \cos(0)\\ &= 1.\checkmark\end{aligned}$$

Hence the sequence (b_k) defined by $b_k = (-1)^k \cos(1/k)$ is divergent and by the divergence test the series $\sum_{k=1}^{\infty} b_k$ is divergent. ✓

4.(a) Let $f : D \to \mathbb{R}$. Then f is said to be differentiable at the point $c \in D$ if the limit

$$f'(c) := \lim_{x\to c}\frac{f(x)-f(c)}{x-c}$$

exists. The number $f'(c)$ is called the derivative of f at c. ✓✓✓

(b)

Proof. A short calculation shows that

$$\frac{f(x)g(x) - f(c)g(c)}{x - c} = \frac{f(x)g(x) + f(x)g(c) - f(x)g(c) - f(c)g(c)}{x - c}.$$

Utilising this expression, the algebra of limits and the fact that differentiable functions are continuous, it follows that

$$\begin{aligned}\lim_{x\to c}\left(\frac{f(x)g(x) - f(c)g(c)}{x - c}\right) &= \lim_{x\to c}(f(x)) \lim_{x\to c}\left(\frac{g(x) - g(c)}{x - c}\right) \\ &\quad + g(c) \lim_{x\to c}\left(\frac{f(x) - f(c)}{x - c}\right) \\ &= f(c)g'(c) + g(c)f'(c).\end{aligned}$$

□

(c)

Proof. Given $\varepsilon > 0$, let $\delta = \varepsilon/3$. Then if $0 < |x - 3/2| < \delta$ it follows that

$$\begin{aligned}|f(x) - 13/2| &= \left|\frac{6x^2 - 5x - 6}{2x - 3} - 13/2\right| \\ &= |3x + 2 - 13/2| \\ &= 3|x - 3/2| \\ &< 3\delta \\ &< \varepsilon.\end{aligned}$$

Hence $\lim_{x\to 3/2}(f(x)) = f(3/2)$. □

(d) This statement is false. The standard example is the modulus function. That is,

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

This function is continuous everywhere but not differentiable at the point $c = 0$.